DOMAIN 5 - ENVIRONMENTAL MANAGEMENT

Q1906 What is the first thing to do when a water sample comes into a lab?

1. Refrigerate the sample
2. Take the pH of the sample
3. Log in and assure chain of custody of the sample
4. Analyze the sample

Ans(3). Log in the sample. Good laboratory procedures dictate that the sample be immediately logged in as soon as possible after arriving at the lab.

Q1907 What is the major factor in determining the required sample volume?

1. Analytical sensitivity
2. Flow rate
3. Type of contaminant
4. Type of air mover

Ans(1). The analytical sensitivity is the determining factor when calculating the required sample volume.

Q1908 Which of the following is an example of a typical VOC?

1. Vinyl chloride
2. Arsenic
3. Aldrin
4. Carbonic Acid

Ans(1). Vinyl Chloride is a typical Volatile Organic Compound (V.O.C.) found in contaminated water. Arsenic is a metal, Aldrin I s pesticide and carbonic acid is an inorganic compound. Volatile organic compounds (VOC) generally means any compound of carbon which participates in atmospheric photochemical reactions, excluding carbon monoxide, carbon dioxide, carbonic acid, metallic carbides or carbonates, and ammonium carbonate.

Q1909 Why is vapor density important is assessing the hazard of a spilled chemical?

1. The concentration of the vapor will determine the hazard
2. The vapor can cause visual problems for responders
3. Vapor density is not an issue in assessing hazard
4. It will determine the behavior of the vapor at the spill scene

Ans(4). Vapor density will determine if the vapor will sink into low areas or dissipate.

Q1910Water is poured into a cylindrical tank at the rate of 9 in^3/min. How much time is required to fill the tank if its radius is 3 in. and its height is 14 in.?

1. 41 min
2. 44 min
3. 47 min
4. 50 min

Ans(2). The volume of a cylinder = πr^2 height
$V = \pi (3 \text{ in})^2 14 \text{ in} = 396 \text{ in}^3$.
Since the rate of flow is 9 in^3/minute, the time required to fill the tank is (396 in^3)/(9 in^3/min) = 44 min

Q1911 A wind rose is:

1. A kind of flower
2. A graphical representation of yearly wind direction
3. A computer-aided design
4. A graphical representation of stack discharges

Ans(2). Wind roses are graphical representations of wind direction. They are important in determining prevailing wind direction

Q1912 In reviewing designs for product safety, all of the following environmental consideration(s) must be dealt with EXCEPT:

1. Environmental conditions on product operations
2. Product operations on the environment
3. Product disposal on the environment
4. Aesthetic design

Ans(4). The environment must be considered in designs for product safety because:

- the ambient environment (temperature, humidity, corrosivity) will affect the condition and operation of the product
- the operation of the product (noise, particles, toxic vapors, liquid effluent) would, without proper controls, pollute the environment
- product disposal, should it contain hazardous materials (mercury, lead, beryllium) has the potential to contaminate soil and groundwater. Aesthetics are not part of a product safety environmental assessment.

Q1913 Elements of a hazardous waste manifest include all of the following EXCEPT:

1. Liability insurance company
2. Manifest number
3. EPA identification number
4. Total quantity of waste generated

Ans(1).A manifest does not require an insurance company number. It does require manifest number, EPA ID number, and total quantity of waste.

Q1914 Which of the following common law causes of action is NOT used to pursue legal remedies in hazardous waste/hazardous materials, or "toxic tort", damage suits?

1. Negligence
2. Nuisance and trespass
3. Strict liability for ultra-hazardous activities
4. Corpus delecti

Ans(4). Negligence, nuisance and trespass, and strict liability for ultra-hazardous activities are legal instruments that are available to plaintiffs in toxic tort suits for damages from hazardous materials/wastes releases or exposure.

Q1915 A Phase I environmental site assessment typically includes the following activities EXCEPT:

1. Inspect the property, review environment records, and assess chemical handling and disposal practices
2. Aerial photo search and review of past ownership
3. Determine whether the subject or surrounding properties are identified on any government lists of contaminated sites, or lists of sites requiring investigation, clean-up, and/or enforcement.
4. Aerial hydrogeologic analysis

Ans(4). Phase I environmental site assessments, a common term for the initial investigation in industrial property transfers, includes the listed activities and others as well, except for aerial analysis of the hydrogeology.

Q1916 Phase I environmental assessments should include, but are not limited to:

1. Historical review of the uses(s) and improvement(s) made to the subject site
2. An investigation of subject and neighboring properties with respect to environmental enforcement actions and/or listing on federal or state contaminated property lists
3. Evaluation of whether present or past owners or tenants have managed hazardous materials or wastes on-site
4. All of the above

Ans(4).Phase I environmental site assessments, as state of the art is currently defined, should include each of the above listed activities.

Q1917 In the RCRA program, UTS refers to:

1. Underlying treatment standards
2. Universal treatment standards
3. Universal technology standards
4. Universal technical standards

Ans(2). Universal treatment standards are one of the key regulatory reforms of the Land Disposal Restrictions program found at 40 CFR 268. The universal treatment standards are the regulatory mechanism whereby a single ("universal") maximum allowable concentration is established for each regulated constituent in wastes that are hazardous at the point of generation, regardless of the waste code(s) carried by the waste. In other words, the concentration limit for the constituent chromium for land disposal purposes would be the same regardless of whether the waste was F006 or K051. Distinctions are still made between wastewater and non-wastewater forms of particular waste codes. The universal treatment standards were introduced in the so-called Phase II LDR final rule, appearing in the September 19, 1994 Federal Register, at 59 FR 47982-48110. The UTS are found at 40 CFR 268.48, appropriately entitled "Table UTS - Universal Treatment Standards". The Phase II rule went into effect on December 19, 1994.

Q1918 A TLV-TWA is defined as:

1. A time-weighted average concentration for a normal 8-hour workday and 40 hour work week to which all workers can be exposed continuously, day after day, without effect.
2. A time-weighted average concentration for a normal 8-hour workday and 40-hour work week, to which nearly all workers can be repeatedly exposed, day after day, without adverse effect
3. A relative index of toxicity
4. Both 2 and 3

Ans(2). TLV-TWA is a time weighted average concentration for a normal 8-hour workday and 40-hour work week, to which "nearly" all workers can be repeatedly exposed, day after day, without "adverse" effect.

Q1919 TLV-STEL is the concentration to which workers can be exposed for a short period of time without suffering from the following, EXCEPT:

1. Dermatitis
2. Irritation
3. Chronic or irreversible tissue damage
4. Narcosis

Ans(1). Dermatitis is not a factor in definition of the TLV-STEL, and therefore, employees may suffer from some dermatitis. A STEL exposure should NOT result in irritation, chronic or irreversible tissue damage, and narcosis of sufficient degree to increase the likelihood of accidental injury, impair self-rescue, or materially reduce work efficiency (provided that the 8-hour TWA-TLV was not exceeded).

Q1920 A STEL is defined as a ____ minute TWA exposure which should not be exceeded at any time during the workday.

1. 1
2. 5
3. 10
4. 15

Ans(4). Exposures above the TLV-TWA up to the short term exposure limit (STEL) should be no longer than 15 minutes. The 15-minute STEL is a time weighted average exposure which should not be exceeded even if the 8-hour TWA is within the TLV- TWA.

Q1921 Exposures up to the TLV-STEL should not be longer than 15 minutes and should not occur more than:

1. Once per day
2. Twice per day
3. Four times per day
4. Eight times per day

Ans(3). Employees should not be exposed to the 15 minute STEL more than four times per day. There should be at least an hour between successive exposures.

Q1922 TLV-C stands for:

1. Threshold Limit Value-Concentration
2. Threshold Limit Value-Ceiling
3. Threshold Limit Value-Carcinogen
4. Threshold Limit Value-Corrected

Ans(2). The Threshold Limit Value-Ceiling is the concentration that should not be exceeded during any part of the working exposure period.

Q1923 Excursions in worker exposure levels may exceed __ times the TLV-TWA for no more than __ minutes during a workday.

1. 1, 10
2. 2, 20
3. 3, 30
4. 5, 50

Ans(3). Excursions in worker exposure levels may exceed 3 times the TLV-TWA for no more than 30 minutes during a workday. This is based upon a geometric standard deviation of 2.0, in which 5% of all values will exceed 3.13 times the geometric mean.

Q1924 Hazardous waste determinations are to be performed at what point in time?

1. Prior to seeking off-site disposal approvals
2. After any on-site treatment activities
3. At the point of generation
4. None of the above

Ans(3). The EPA, in both recent rulemakings regarding listings and characteristics of hazardous wastes, as well as in the Land Disposal Restrictions rulemakings, has made their position regarding the importance of point of generation determinations quite clear. The determination may need to be redone after any on-site treatment, but such treatment does not alter the need for hazardous waste determinations to be performed at the point of generation for any wastes undergoing treatment on-site.

Q1925 The EPA considers all of the following pollutants "criteria pollutants" EXCEPT:

1. Carbon monoxide
2. Oxides of nitrogen
3. Oxides of sulfur
4. Hydrogen sulfide

Ans(4). Hydrogen sulfide is not a criteria pollutant.

Q1926 Acronyms are used extensively in the various regulatory programs as a form of shorthand. Which of the following acronyms is NOT associated with the CWA program?

1. POTW (Publicly Owned Treatment Works)
2. SPCC (Spill Prevention Control and Counter Measures)
3. NAAQS (National Ambient Air Quality Standards)
4. BAT (Best Available Treatment)

Ans(3). National Ambient Air Quality Standards (NAAQS) is associated with the Clean Air Act program. POTW, SPCC, and BAT are all terms associated with the Clean Water Act program.

Q1927 The "Superfund" law primarily addresses:

1. Worker safety
2. Toxic waste sites
3. PCB's
4. Carcinogens

Ans(2). The Superfund law was written to force clean-up of sites contaminated with toxic materials. Sites are prioritized and those of highest priority are listed on the NPL (National Priorities List). Money is set aside to clean these sites up if the responsible parties cannot be found or refuse to clean it up.

Q1928 Drivers of any size vehicle transporting hazardous materials that require placarding must pass the pre-trip skills and written test to obtain a:

1. Class I license
2. Class II license
3. CDL
4. Either 1 or 2

Ans(3). Under the terms of the Commercial Motor Vehicle Act, anyone driving a vehicle weighing more than 26,000 pounds, transporting more than 15 people including the driver, or transporting hazardous materials that require placards, must pass the pre-trip skills and written test to obtain a commercial driver's license (CDL).

Q1929 Which of the following agencies has an interest in safety and health matters and regulates selected chemical substances and mixtures?

1. NIOSH
2. EPA
3. DOT
4. NACOSH

Ans(2). The Environmental Protection Agency (EPA) was established by the Toxic Substances Control Act (TSCA) in 1976 to regulate selected chemical substances and mixtures being manufactured, processed, distributed, imported, and disposed of, if they are considered an unreasonable risk to health and the environment.

Q1930 A firm makes a plated device that it installs on the purchaser's property. The device discharges a water effluent. Which of the following agencies govern the device operation?

1. OSHA and NRC
2. EPA and OSHA
3. NRC and DOT
4. EPA and DOT

Ans(2). Normal operation of the device would be governed by EPA and OSHA.

Q1931 The Toxic Substances Control Act (TSCA) is enforced by the:

1. The Toxic Substances Control Act (TSCA) is enforced by the:
2. Environmental Protection Agency
3. Occupational Safety and Health Administration
4. Consumer Product Safety Commission

Ans(2). The EPA enforces TSCA.

Q1932 The key element of RCRA designed to track hazardous waste from "cradle to grave" is the:

1. Statute of limitations
2. Criminal liability
3. Manifest
4. Ultimate disposal site

Ans(3). The manifest system was developed by RCRA to track hazardous waste from originator to final resting place.

Q1933 A manifest serves the following purposes:

1. It is a shipping document for hazardous wastes
2. It is a tracking document for hazardous wastes
3. It is a certificate of destruction for hazardous wastes
4. Both 1 and 2

Ans(4). The manifest serves both shipping paper requirements under DOT regulations, and acts as a tracking document under RCRA regulations. The manifest is not a fail-safe document, however, and does not absolve the generator of CERCLA liabilities. At best the manifest merely indicates that the waste actually was received by the designated facility.

Q1934 The mandatory shipping document for hazardous waste purposes is:

1. Uniform hazardous materials bill of lading
2. Straight bill of lading
3. Uniform hazardous waste manifest
4. Dangerous good declaration

Ans(3). The uniform hazardous waste manifest (EPA Form 8700-22) is the mandatory format for hazardous waste shipping documents. The actual forms are to be obtained from states mandating the use of state issued uniform hazardous waste manifests, or when the state is not involved in manifest distribution, from commercial printers following the mandatory format with the appropriate number of pages (minimum of 4).

Q1935 An industrial wastewater stream has high levels of dissolved copper that must be reduced before discharge. The wastewater is chemically treated to remove the copper. Which of the following will NOT drive the reaction to remove more copper?

1. Removing one of the products as a precipitate
2. Removing one of the products as a gas
3. Reducing the concentration of one of the reactants
4. Removing one of the products as a slightly ionized substance

Ans(3). Reducing reactant concentration tends to retard the chemical reaction process. On the other hand, if a product is removed, the chemical equilibrium is perturbed, and the reactants will continue to react until a new equilibrium is achieved.

Q1936 Which of the following chemicals is not a solvent in the operation described?

1. Alcohols in paint cleanup
2. Ketones in spray painting
3. Hydrocarbons in metal cleaning
4. Epoxies in coremaking in foundries

Ans(4).To be a solvent, the chemical must dissolve something. Epoxies in coremaking form a resin matrix (i.e., a polymer) that binds sand together - no solvent

Q1937 The greenhouse effect is due to a disruption in which geochemical cycle?

1. Nitrogen
2. Sulfur
3. Carbon
4. Oxygen

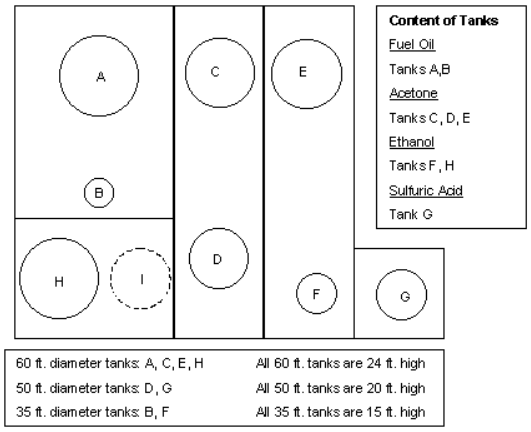
Ans(3). The greenhouse effect is caused by a disruption in the global carbon cycle.

Q1938 In assessing mobility and persistence of hazardous constituents upon release to the environment, the following factors are of interest EXCEPT:

1. Physical state of the matrix
2. Water solubility
3. Octanol-water partition coefficient (Kow) and soil-sorption coefficient (Koc)
4. Electrofluorescence

Ans(4). Electrofluorescence is not a factor considered in mobility and persistence. The other answers are of value in predicting the fate, transport, and persistence of chemicals released to the environment.

Q1939



Materials	Separation Distance
Acetone and ...	
Acetone	50 ft.
Ethanol	50 ft.
Fuel Oil	40 ft.
Sulfuric Acid	80 ft.
Ethanol and ...	
Acetone	50 ft.
Ethanol	50 ft.
Fuel Oil	40 ft.
Sulfuric Acid	80 ft.
Fuel Oil and ...	
Acetone	40 ft.
Ethanol	40 ft.
Fuel Oil	30 ft.
Sulfuric Acid	80 ft.
Sulfuric Acid and ...	
Acetone	80 ft.
Ethanol	80 ft.
Fuel Oil	80 ft.
Sulfuric Acid	55 ft.

tank farm is being designed and will be located in a geographical location that receives extensive rainfall. The tank farm will house four hazardous materials: fuel oil, ethanol, acetone, and sulfuric acid. To ensure that corporate safety and environmental protection policies are followed, you have been asked to provide input on the layout of the containment dikes.
Because of the soil, no dike can be higher than 4.5 ft. Due to the potential for heavy rains, company policy requires a 25% overage be designed into the capacity of any given diked volume.
What are the minimum dimensions of the diked area necessary to protect acetone tanks C and D?

1. 145 ft * 145 ft
2. 130 ft * 160 ft

3. 138 ft * 160 ft
4. 138 ft * 138 ft

Ans(2). Answer this question one step at a time.

Step 1: Find the capacity of the larger tank. Dikes must be able to fully contain a release of the largest tank in the diked area. Tank C is the larger tank. Assume that its capacity is the same as the volume of a cylinder. The diameter of Tank C is 60 ft an d its height is 24 ft. Its volume is therefore
$V = \pi [(60 \text{ ft})^2 / 4] 24 \text{ ft}$
$= 67{,}858 \text{ ft}^3$

Step 2: The next step is to determine the required capacity of the diked area. The dike must contain a complete loss of the largest tank plus a rainfall overage of 25%. Therefore, the actual capacity of the dike is 25% higher than the volume of the largest tan k.
Total Volume = $1.25 * 67{,}858 \text{ ft}^3$
$= 84{,}823 \text{ ft}^3$

Step 3: Unfortunately, we cannot use the volume occupied by the other tank in the diked area (Tank D). We DO assume that the volume occupied by Tank C is available, though. To calculate the unavailable dike volume occupied by Tank D, simply assume that Tank D is a cylinder of 4.5 ft high (the height of the dike) and calculate the volume of this 4.5 ft high cylinder.
Volume Tank D = $\pi[(50 \text{ ft})^2 / 4]4.5 \text{ ft}$
$= 8{,}836 \text{ ft}^3$

Step 4: Calculate the Net Volume. This is the Total Volume plus the Volume of Tank D.
Net Volume = $84{,}823 \text{ ft}^3 + 8{,}836 \text{ ft}^3$
$= 93{,}659 \text{ ft}^3$

Step 5: Calculate the required surface area necessary to cover this volume. We are allowed a dike height of 4.5 ft; therefore, the required surface area is the Net Volume divided by the dike height.
Surface Area = $93{,}659 \text{ ft}^3 / 4.5 \text{ ft}$
$= 20{,}813 \text{ ft}^2$

Step 6: To calculate the preliminary minimum dimensions, simply take the square root of the Surface Area. This will give us the preliminary minimum dimensions in the form of a square.
Minimum Dimension = $\sqrt{20{,}813}$ ft^2
= 144.3 ft * 144.3 ft

Step 7: Verify that these dimensions are compatible with tank separation requirements. Tank C must be at least 50 ft from Tank D. Assume that all other tanks in this tank farm can be placed sufficiently far away from Tank C in their own diked areas to meet minimum separation requirements.
Tank C has a diameter of 60 ft, and Tank D has a diameter of 50 ft. The minimum separation distance is 50 ft. Therefore, when adding these gross dimensions, you find that the vertical dimension (with respect to the graphic) of 144.3 ft is insufficient. If the two tanks were built right on the upper and lower dikes, they would only be separated by 34.3 ft. An additional

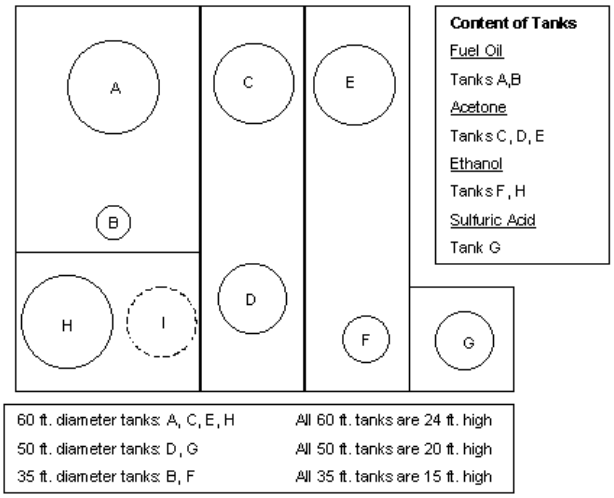
15.7 ft in the vertical dimension is required to account for minimum separation.

Step 8: The minimum vertical dimension is therefore
Vertical Dimension = 144.3 ft + 15.7 ft
= 160 ft

Step 9: Calculate the new horizontal dimension (with respect to the graphic). The Net Volume is still 93,659 ft^3. The new horizontal dimension is therefore:
Horizontal Dimension = 93,659 ft^3 / 160 ft / 4.5 ft
= 130 ft
Finally, the minimum dimensions of this diked area are 160 ft vertically * 130 ft horizontally.

NOTE: Always go through the process of determining the preliminary minimum dimensions. If Tank C and Tank D contained fuel oil instead of acetone, for example, the preliminary minimum dimensions would have been the actual minimum dimensions.

Q1940



Materials	Separation Distance
Acetone and . . .	
Acetone	50 ft.
Ethanol	50 ft.
Fuel Oil	40 ft.
Sulfuric Acid	80 ft.
Ethanol and . . .	
Acetone	50 ft.
Ethanol	50 ft.
Fuel Oil	40 ft.
Sulfuric Acid	80 ft.
Fuel Oil and . . .	
Acetone	40 ft.
Ethanol	40 ft.
Fuel Oil	30 ft.
Sulfuric Acid	80 ft.
Sulfuric Acid and . . .	
Acetone	80 ft.
Ethanol	80 ft.
Fuel Oil	80 ft.
Sulfuric Acid	55 ft.

A tank farm is being designed and will be located in a geographical location that receives extensive rainfall. The tank farm will house four hazardous materials: fuel oil, ethanol, acetone, and sulfuric acid. To ensure that corporate safety and environmental protection policies are followed, you have been asked to provide input on the layout of the containment dikes.
Because of the soil, no dike can be higher than 4.5 ft. Due to the potential for heavy rains, company policy requires a 25% overage be designed into the capacity of any given diked volume.
Because of piping layout requirements, the edge of Tank F will be located 1 ft from the dike it shares with Tank G. The same piping issues require the centerlines of Tank F and Tank G to line up.
What are the minimum dimensions of the other three dikes containing Tank G? Assume that the diked area for Tank G is rectangular.

1. 85 ft * 85 ft
2. 105 ft * 85 ft
3. 105 ft *105 ft
4. 129 ft * 85 ft

Ans(4).Answer this question one step at a time.

Step 1: Find the capacity of Tank G. Assume that its capacity is the same as the volume of a cylinder. The diameter of Tank G is 50 ft and its height is 20 ft. The volume is therefore:
V = □ [(50 ft)^2 / 4] 20 ft
= 39,270 ft^3

Step 2: The next step is to determine the required capacity of the diked area. The dike must contain a complete loss of Tank G plus a rainfall overage of 25%. Therefore, the actual capacity of the dike is 25% higher than the volume of Tank G.
Total Volume = 1.25 * 39,270 ft^3
= 49,087 ft^3

Step 3: Since there are no other tanks in this diked area, we can immediately calculate the required surface area to cover this volume by dividing the volume by the dike height of 4.5 ft.
Surface Area = 49,087 ft^3 / 4.5 ft
= 10,908 ft^2

Step 4: To calculate the preliminary minimum dimensions, simply take the square root of the Surface Area. This will give us the preliminary minimum dimensions in the form of a square.
Minimum Dimension = $\sqrt{10,908}$ ft^2
= 104.4 ft * 104.4 ft

Step 5: Verify that these dimensions are compatible with the tank separation requirements. Tank G must be at least 80 ft from Tank F. Assume that Tank E is sufficiently far away from Tank G to meet minimum separation requirements. Tank G has a diameter of 50 ft. If it were placed at the far end of its diked area, it would still be too close to Tank F by 24.6 ft (104.4 ft - 50 ft - 80 ft + 1 ft). (The 1 ft dimension is the distance of Tank F from the shared dike.)
Therefore, the minimum horizontal dimension (with respect to the graphic) is
Minimum Horizontal Dim. = 104.4 + 24.6 ft = 129 ft

Step 6: Calculate the new minimum vertical dimension (with respect to the graphic) based on the minimum horizontal dimension of 129 ft using the Total Volume.
Minimum Vertical Dim. = 49,087 ft^3 / 129 ft / 4.5 ft = 84.6 ft

NOTE: Always go through the process of determining the preliminary minimum dimensions. If Tank G contained any of the other three materials in this exercise, the preliminary minimum dimensions would have been the actual minimum

Q1941 All of the following agents can contribute to indoor air pollution EXCEPT:

1. Carbon dioxide
2. Formaldehyde
3. Asbestos
4. PCB's

Ans(4). Although PCB's have caused problems in building contamination, they have a low vapor pressure and do not cause indoor air quality problems, unless burned. Carbon dioxide, formaldehyde, and asbestos are common indoor air pollutants that are caused by building materials and occupants.

Q1942 Which of the following sources of indoor air pollution in an office environment can be lethal?

1. Microorganisms
2. Formaldehyde
3. Carbon dioxide
4. Poor ventilation

Ans(1). Airborne microorganisms circulated from water-cooling systems (Legionella Pneumophila) have caused Legionnaires Disease from acute exposures. Formaldehyde and carbon dioxide require concentrations well beyond those found in offices to be lethal. Poor ventilation would require reductions in oxygen level below 19.5%, which is unlikely in an office environment.

Q1943 Methods to control Legionnaires disease include the following EXCEPT :

1. Continuous feed biocide
2. Locate make-up air intakes upwind from cooling towers
3. Periodic cleaning of cooling water systems
4. Sampling to ensure bacterial control

Ans(1). Continuous feed biocides can cause resistant strains of bacteria in the cooling system. It is best to alternate biocides to prevent resistance. Prevention of cooling tower aerosols into the building makeup air will prevent bacteria from entering the ventilation system. Complete periodic cleaning and sampling of the cooling water systems have proven effective.

Q1944 What capacity of outdoor (fresh) air does the American Society of Heating, Refrigerating, and Air Conditioning Engineers (ASHRAE) recommend in its 62-1989 Standard for an office environment?

1. 10 cfm/person
2. 20 cfm/person
3. 10 cfm/room
4. 20 cfm/room

Ans(2).ASHRAE, in its 62-1989 Standard, recommends 20 cfm/person of outdoor air for acceptable indoor air quality in an office environment. This value has been chosen to control carbon dioxide and other contaminants with an adequate margin of safety and to account for health variations among people, varied activity levels, and a moderate amount of smoking.

Q1945 Secondary containment should be considered for all EXCEPT:

1. 90-day container accumulation areas
2. Chemical and waste loading and unloading areas
3. Virgin product storage areas and tanks
4. Immersion heaters

Ans(4). Secondary containment of chemical and waste storage and transfer areas is a best management practice and should be seriously evaluated even where not currently required by federal, state, or local regulations. Immersion heaters present a fire hazard in electroplating operations but do not require secondary containment

Q1946 Secondary containment is considered:

1. A best management practice
2. Necessary to effective wastewater treatment
3. Provided by compacted soil barriers
4. Unnecessary if hazardous waste tanks and containers are frequently inspected and when releases are directed away from storm sewers

Ans(1). Regardless of regulatory requirements, secondary containment is recommended as a best management practice. Secondary containment manages the risks and costs associated with chemical releases. Requirements for secondary containment can be found in both RCRA and CWA regulations. The fines, clean up costs, and litigation expenses associated with oil and chemical releases indicate the necessity for secondary containment.

Q1947 How is asbestos disposed of?

1. Landfill
2. Incinerated
3. Chemically dissolved
4. Deep well injection

Ans(1). Asbestos waste is commonly landfilled.

Q1948 What type of facility would be least likely to generate lab packs?

1. University
2. Semiconductor manufacturing facility
3. Chemistry lab
4. Hospital

Ans(2). Except for semiconductor manufacturing facility, all of the above facilities would use a wide variety of small quantities of chemicals and would then require lab packs.

Q1949 Why are temperature inversions important in air pollution?

1. They serve to disperse pollutants
2. They serve to concentrate pollutants
3. They are not important
4. They increase reaction of the chemicals with ozone

Ans(2). Temperature inversions serve to concentrate air pollutants near the ground.

Q1950 The level of radon exposure in a home considered by the EPA to be the maximum acceptable level is:

1. 0.4 picocuries/m^3
2. 4.0 picocuries/m^3
3. 0.4 picocuries/l
4. 4.0 picocuries/l

Ans(4). The EPA has recommended the level of 4.0 picocuries/l air as a maximum safe level of exposure to radon in the home environment.

Q1951 Maximum contaminant levels (MCL's) are regulatory thresholds for:

1. Effluent quality under the Clean Water Act
2. Air emissions under the Clean Air Act
3. Heavy metal contamination under RCRA
4. Allowable drinking water pollutant concentrations under the Safe Drinking Water Act

Ans(4). MCL's represent maximum permissible levels of specified contaminants in water which is delivered to any user of a public water supply system.

Q1952 Primary MCL's refer to:

1. Concentrations of regulated drinking water contaminants that are protective of human health with a margin of safety
2. Groundwater quality standards
3. Maximum levels of contaminants allowable in public water systems
4. Workplace exposure limits for airborne contaminants

Ans(3). Primary maximum contaminant levels (MCL's) are the maximum permissible levels of specific contaminants in water which is delivered to any user of a public water supply system. Answer 1 refers to maximum contaminant level goals (MCLG's). MCL's are sometimes used as groundwater cleanup goals in the RCRA and CERCLA programs.

Q1953 MCLG's refer to:

1. Enforceable drinking water quality standards
2. Non-enforceable concentrations of drinking water contaminants that are protective of health with a margin of safety
3. Non-enforceable standards addressing the aesthetic qualities of drinking water, such as appearance, odor, and taste
4. Regulatory standards for small public water supply systems

Ans(2). Maximum contaminant level goals (MCLG's) under the Safe Drinking Water Act (SDWA) are non-enforceable health-based standards, and are used in developing legally enforceable MCL's. Aesthetic qualities of drinking water are addressed by non-enforceable secondary MCL's.

Q1954 Industrial dischargers to a POTW should be concerned when a POTW violates its NPDES permit because:

1. Potential bans on new sewer connections
2. Potential additional or more stringent retreatment requirements
3. Potential liability for contamination of sewage treatment plant sludge
4. All of the above

Ans(4). Industrial dischargers to the sewer to a publicly owned treatment works (sewage treatment plant) should be very concerned when the POTW violates its NPDES permit for all of the above listed reasons, as well as for the real potential for increased scrutiny of its own operations by federal, state, and local officials, and environmental groups. Increased enforcement activity and citizen suits are likely to follow.

Q1955 Chemical analysis of the products of combustion will almost always show the presence of what element in the molecules?

1. Silica
2. Hydrogen
3. Carbon
4. Oxygen

Ans(4).Since oxygen is required to support normal combustion, the products of combustion would contain oxygen in combination with other elements.

Q1956 Which of the following general classes of insecticides is the most persistent in the environment?

1. Carbamate
2. Organochlorine
3. Organophosphate
4. Pyrethrins

Ans(2). The carbamate and organophosphate insecticides are degraded relatively rapidly in both the environment and the mammalian body compared to the organochlorine insecticides. Organochlorine pesticides are relatively persistent in the environment. DDT was an organochlorine pesticide

Q1957 What type of emergencies would you expect to find at a TSD site?

1. Those involving hazardous substances
2. Those involving trash or rubbish
3. Those dealing with fire and similar life threatening conditions
4. Those involving gas cylinders

Ans(1). Treatment, storage, and disposal (TSD) sites are required by OSHA, TSCA, and CERCLA to have an emergency response plan for all emergencies involving hazardous substances. OSHA's HAZWOPER Standard, 29 CFR 1910.120, is the standard that details program requirements.

Q1958 All of the following EXCEPT _____ are components of photochemical air pollution.

1. Ozone
2. Sulfur dioxide
3. Hydrocarbons
4. Oxides of nitrogen

Ans(2). Photochemical air pollution arises from a series of atmospheric reactions - NO_2 absorbs UV light giving rise to NO and O radicals which in turn react to produce O_3 (ozone). Ozone and NO can react to regenerate NO_2. Thus the reaction is cyclic. Sulfur dioxide is not part of the photochemical smog problem.

Q1959 Beer's Law refers to:

1. Spectrophotometry
2. Chromatography
3. Dosimetry
4. Fluorescence

Ans(1). Beer's Law is A = kC, where A is the absorbance of a solution, C is the concentration of analyte, and k is a constant whose value depends on the wavelength of the radiation, the nature of absorbing system, and the optical path length, or cell thickness.

Q1960 The phenomena of biomagnification of heavy metals is due to the:

1. Second Law of Thermodynamics
2. Lead pipes
3. Collapsed sewer mains
4. Food chain

Ans(4). The food chain serves to concentrate heavy metals in the uppermost predators as the lower forms add to their heavy metal loading.

Q1961 What does BOD stand for?

1. Biological Organic Demand
2. Basic Oxygen Demand
3. Biochemical Oxygen Demand
4. Biological Oxidation Demand

Ans(3). BOD stands for Biochemical Oxygen Demand. This is the quantity of oxygen used in the aerobic stabilization of wastewaters.

Q1962 A cyclone scrubber is put on-line with an Electrostatic Precipitator (ESP). The efficiency of the cyclone is 80% while the ESP is 95%. Assuming the cyclone is last, what is the overall system efficiency?

1. 99%
2. 99.9%
3. 99.99%
4. 99.999%

Ans(1). Efficiency = 95% + (80 * 5%)
= 99% efficiency

Q1963 A fluid system with a Reynolds number significantly below 2,000 implies what about a fluid system's flow characteristics?

1. Laminar flow
2. Turbulent flow
3. Too fast for dust transport
4. Too slow for dust transport

Ans(1). The Reynolds number is a dimensionless number used to model whether the flow is turbulent or laminar. If the Reynolds number is less than 2,000 the flow will have laminar characteristics. If it is greater than 2,000 the fluid system is considered turbulent .

Q1964 A liquid with a pH of 1 is:

1. Basic
2. Acidic
3. Neutral
4. Alkaline

Ans(2). A liquid with a pH of 1 is highly acidic. Basic and alkaline are synonyms of a liquid with a pH greater than 7. A liquid with a pH of 7 is neutral.

Q1965 A chemical compound capable of forming multiple chemical bonds to a metal ion is called a:

1. Chelating agent
2. Reactive agent
3. Catalyst
4. Reactive ion

Ans(1). A chelating agent is a chemical compound capable of forming multiple chemical bonds to a metal ion.

Q1966 A substance that affects the change in hydrogen ion concentration (pH) otherwise produced by adding acids or bases to a solution is called a:

1. Buffer
2. Bonding agent
3. Catalyst
4. Chloracne agent

Ans(1). A buffer is a substance that reduces the change in hydrogen ion concentration (pH) otherwise produced by adding acids or bases to a solution.

Q1967 Which one of the following correctly balances the following reaction: NaOH + H_2SO_4?

1. $Na_2SO_5 + H_2O$
2. $SO_5H_2O + Na_2$
3. $Na_2SO_3 + H_2O_2$
4. $Na_2SO_4 + 2\ (H_2O)$

Ans(4).This is an example of an acid base neutralization reaction. The resulting salt from this reaction is sodium sulfate. To properly balance the full reaction equation, two sodium hydroxide molecules react with one sulfuric acid molecule to yield $Na_2SO_4 + 2\ (H_2O)$.

Q1968 All the following are true about ketones EXCEPT:

1. Ketones have one or more carbonyl groups (C = O)
2. Methanol, a small-chain ketone, can cause blindness
3. Long term exposure to ketones can adversely affect the brain and nervous system
4. Ketones are commonly used as solvents

Ans(2). Methanol is not a ketone; it is an alcohol.

Q1969 Which one of the following equations is NOT balanced?

1. $2\ (Mg) + O_2 \rightarrow 2\ (MgO)$
2. $4\ (Fe) + 3\ (O_2) \rightarrow 2\ (Fe_2O_3)$
3. $H_2O + N_2O_3 \rightarrow 2\ (HNO_2)$
4. $2\ (Na_2O) + H_2O \rightarrow 2\ (NaOH)$

Ans(4). Only one molecule of Na_2O reacts with one molecule of water to produce 2 molecules of NaOH.

Q1970 25 grams of H_20 is equivalent to how many moles of H_2O?

1. 0.72 mol
2. 1.00 mol
3. 1.39 mol
4. 25.0 mol

Ans(3). mole is the equivalent of one gram-molecular weight. Since the molecular weight of water is 18 g/mol, 25 grams is equivalent to 25 g / (18 g/mol) = 1.39 mol.

Q1971 What volume will 38 g of oxygen occupy at 1 atm and 20ºC?

1. 22.4 liters
2. 28.5 liters
3. 29.3 liters
4. 32.6 liters

Ans(2). PV = nRT
R = 0.08205 l-atm/mol-K because we are interested in liters with pressure in atm and temperature in K. Remember to use absolute temperatures. The gram-molecular weight of oxygen (O_2) is 32 g/mol.
V = nRT / P
= [(38 g/32 g/mol)0.08205 l-atm/mol-K * 293° K)] / 1 atm
= 28.5 liters

Q1972 What volume does 1 mole of ANY gas occupy at 1 atm and 0ºC?

1. 22.4 liters
2. 1 liter
3. 24.45 liters
4. 1 ft^3

Ans(1). PV = nRT
V = nRT / P
= (1 mol * 0.08205 l-atm/mol-K * 273° K)
= 22.4 liters

Q1973 One atmosphere is equivalent to how many pounds per square inch pressure?

1. 1 psi
2. 14.7 psi
3. 29.92 psi
4. 760 psi

Ans(2). One atmosphere is equivalent to 14.7 pounds per square inch pressure.

Q1974 What is the partial pressure of a vapor with an airborne concentration of 150 ppm (assume barometric pressure of 1 atmosphere)?

1. 150×10^{-4} atm
2. 1.5×10^{-4} atm
3. 0.15×10^{-4} atm
4. 1.5×10^{-3} atm

Ans(2). Use the formula to calculate the concentration of a vapor and solve for the partial pressure.
$C = (Pv \times 10^{6}) / Pb$ 150
$ppm = (Pv \times 10^{6}) / 1$ atm
$Pv = 1.5 \times 10^{-4}$ atm

Q1975 Convert a partial pressure of 1.5×10^{-4} atm to mm Hg.

1. 4.5×10^{-3} mm Hg
2. 0.114 mm Hg
3. 5.1×10^{-3} mm Hg
4. 0.45×10^{-3} mm Hg

Ans(2). 1 atm = 760 mm Hg
mm Hg = 1.5×10^{-4} atm (760 mm Hg / 1 atm)
= 0.114 mm Hg

Q1976 The most prevalent all-around solvent is:

1. Alcohol
2. Aliphatic hydrocarbons
3. Halogenated hydrocarbons
4. Water

Ans(4). Water is the most prevalent all-around solvent. It is nontoxic, nonflammable, and with proper additives, forms a solvent system that works well even with many organic materials.

Q1977 A fiber is defined as having an aspect ratio (length to diameter) greater than:

1. 5 to 1
2. 2 to 1
3. 3 to 1
4. 10 to 1

Ans(3). Fibers are particles which have a length that is at least three times its width.

Q1978 Gas chromatography (GC) is an excellent separation technique for many organic compounds found in the industrial environment. Which of the following is often performed after GC to identify the compound?

1. Infrared spectrophotometry
2. Mass spectrometry (MS)
3. High performance liquid chromatography (HPLC)
4. Atomic absorption

Ans(2). GC/MS is commonly performed to separate and identify the compound of interest. GC separates a mixture based on differences in boiling point and solubility. Differences in these properties produce differences in retention times in the GC column separating the mixture. MS fractures and ionizes each compound into a recognizable pattern.

Q1979 The conversion of ammoniacal nitrogen to nitrate ion through the aeration of activated sludge is called:

1. Nitrogen fixation
2. Nitrification
3. Nitrate reduction
4. Denitrification

Ans(2). In the sewage treatment process, nitrification converts ammoniacal nitrogen to nitrate ion during the aeration of waste water.

Q1980 Two nuclei of the same element that have the same charge but different masses are called:

1. Neutrons
2. Reactions
3. Protons
4. Isotopes

Ans(4). Isotopes of the same element have the same charge but different nuclear masses.

Q1981 Normally formless fluids that occupy the space of enclosure and which can be changed to the liquid or solid state by increased pressure and decreased temperature are called:

1. Smoke
2. Mists
3. Gases
4. Vapors

Ans(3). Gases are normally formless fluids that occupy the space of enclosure and can be changed to the liquid or solid state by increased pressure and decreased temperature.

Q1982 Which aliphatic hydrocarbon features double carbon bonds?

1. Alkanes
2. Alkenes
3. Alkynes
4. Paraffins

Ans(2).Alkenes (olefins) are unsaturated hydrocarbons with one or more double bonds. Alkanes are unsaturated hydrocarbons with only single carbon bonds. Alkynes are unsaturated hydrocarbons with one or more triple bonds.

Q1983 The tendency of a liquid to vaporize at room temperature is its:

1. Rate of evaporation
2. Vapor pressure
3. Volatility
4. Flammable range

Ans(3). Volatility is the tendency or ability of a liquid to vaporize. Ethanol and gasoline are examples of volatile liquids because of their tendency to evaporate rapidly.

Q1984 The latent heat of vaporization of water is:

1. 1,200 Btu/lb
2. 970 Btu/lb
3. 560 Btu/gal
4. 600 Btu/gal

Ans(2). 970 Btu/lb is the latent heat of vaporization of water. It takes a relatively large amount of heat to vaporize water.

Q1985 A chemical breakdown due to heat is known as a:

1. Chain reaction
2. Decomposition reaction
3. Pyrolysis reaction
4. Pyrotechnic reaction

Ans(3). Pyrolysis is the breakdown of materials due to heat.

Q1986 The pressure at which a vapor is at equilibrium is called the:

1. Specific gravity
2. Molecular weight
3. Boiling point
4. Vapor pressure

Ans(4). Vapor pressure is the pressure at which molecules escaping from a liquid are equal to the number of molecules returning to the liquid from the vapor.

Q1987 In the last 15 years, air pollution emissions have:

1. Increased slightly
2. Doubled
3. Stayed the same
4. Decreased

Ans(4). In the last 15 years many air pollutants, like automobile emissions, have been substantially reduced. For example, an automobile purchased in 1990 emitted approximately 4% of that of a similar 1970 model. Another example is sulfur dioxide, the main ingredient of acid rain that is emitted from coal-burning electric utilities. Compared with 1977, the emitted rate of SO_2 is now 20% lower. Much of this can be attributed to regulations that require controls like after-burners and scrubbers.

Q1988 The greatest single contributor to urban air pollution is:

1. Electric utilities burning oil
2. Electric utilities burning coal
3. Automobiles
4. Process manufacturing plants

Ans(3). Cars and trucks are major sources of carbon monoxide, nitrogen oxides, and volatile organic compounds that interact in the presence of sunlight to produce ground-level ozone pollution commonly referred to as smog. Researchers allege the automobile is responsible for approximately 40% of urban smog nationally and a great deal more in some areas of California.

Q1989 Background concentrations in contaminated soil investigations refer to:

1. Constituent concentrations in samples from areas not affected by the activity being monitored
2. The maximum allowable incremental contamination allowed by EPA
3. Clean-up standards necessary to protect public health and environmental quality
4. Statistical evaluation of data from the potentially contaminated area

Ans(1). Background concentrations typically are of concern only with respect to those constituents naturally occurring in soil. Many regulatory officials believe Answers 2 and 3 to be true. However, background concentrations of a variety of chemical constituents vary widely even in uncontaminated soils. Background concentrations are often used in statistical comparisons of levels in areas under investigation to those that are presumably uncontaminated.

Q1990 The term for the interaction of two toxins where the combined effect is greater than the sum is:

1. Additive
2. Synergistic
3. Potentiation
4. Antagonistic

Ans(2). Synergistic effects occur when two substances act together to produce a toxic effect greater than the sum of each. For example, asbestos exposure and cigarette smoking produce a cancer risk that is much greater than the sum of each risk. Additive effects are the sum of each substance on the body. Potentiation is where one substance has no effect unless combined with a second substance, and antagonistic effects occur when two substances interfere with each others effects, giving an overall lesser effect.

Q1991 Which of the following is the second most plentiful element in the earth's crust:

1. Aluminum
2. Oxygen
3. Iron
4. Silicon

Ans(4). The second most plentiful element in the earth's crust is silicon.

Q1992 Failure to adequately clean-up chemical releases can result in:

1. The spill area being considered an unpermitted land disposal unit
2. Extensive soil and ground water contamination
3. Enforcement actions
4. All of the above

Ans(4). All of the above listed consequences can result from the failure to adequately remediate chemical releases. Another impact is a potential decrease in property value.

Q1993 All of the following are natural sources of air pollution EXCEPT:

1. Volcanoes
2. Soil
3. Vegetation
4. Fossil fuels

Ans(4). Fossil fuel is a component of the global carbon cycle that man mines and combusts; therefore, it is a man-made pollution source.

Q1994 Diesel engines produce ____ pollutants than do gasoline engines:

1. Less
2. More
3. Same amount of
4. Less if compression ratios are higher than gasoline engines

Ans(2). Diesel engines produce more pollutants than do comparable gasoline engines. Compression ratios are necessarily higher in diesel engines.

Q1995 Viable reasons for incineration of air pollutants include all of the following EXCEPT:

1. Odor control
2. Reduce VOC emissions
3. Reduce explosion hazard
4. Reduce ozone emissions

Ans(4).Incineration reduces odors, effectively removes most VOC's, and reduces explosion hazards at plants by not allowing build up of flammables.

Q1996 Based on the tonnage of material emitted annually, which of the following sources accounts for the most air pollution in the United States?

1. Industry
2. Electric power generation
3. Refuse disposal
4. Transportation

Ans(4).Transportation, including private automobiles, accounts for approximately 60% of the air pollution in the United States.

Q1997 Atomic absorption is distinctly good for what type of analysis?

1. Metals
2. Solvents
3. Aromatic hydrocarbons
4. Aliphatic hydrocarbons

Ans(1).Atomic absorption is an analytical technique which takes advantage of the characteristic absorption by metals of certain wavelengths of light.

Q1998 Convert 12 ⌈g/m^3 of phosgene gas to ppm.
Formula for phosgene is $COCl_2$
Atomic weights: Cl = 35.5, O = 16, C = 12

1. 0.003 ppm
2. 0.05 ppm
3. 3.00 ppm
4. 30.0 ppm

Ans(1). The general formula for conversion is given below:
ppm = [(mg/m^3) 24.4] / molecular weight
There are 1,000 ⌈g in 1 mg ppm
= 0.012 mg/m^3 * 24.4 / 99
= 0.003 ppm

Q1999 Convert 400 ppm of sulfur dioxide vapors to percent by volume.
The molecular weight of SO_2 is 64:

1. 4%
2. 0.4%
3. 0.04%
4. 0.004%

Ans(3). 400 ppm means that there are 400 parts of sulfur dioxide for every million parts of the gas.
(400 parts /1,000,000 parts)100% = 0.04%

Q2000 If a vessel contains 80% air and 20% hydrocarbons, what is the oxygen content of the vessel?

1. 21.9%
2. 16.8%
3. 15.5%
4. 14.0%

Ans(2). Normal air contains 21% oxygen by volume, and the vessel contains 80% normal air.
Oxygen content = 0.80 * 0.21
= 0.168
= 16.8%

Q2001 In a gaseous mixture at 20ºC, the partial pressure of the component gases are:
H_2 = 200 mm Hg
CO_2 = 135 mm Hg
N_2 = 320 mm Hg
Ar = 105 mm Hg

What is the volume percent of H_2?

1. 20.0%
2. 23.6%
3. 26.3%
4. 95.0%

Ans(3). P(total) = (200 + 135 + 320 + 105) mm Hg
= 760 mm Hg
Fractional volume of H_2 = P(H_2) / P(total)
= 200 mm Hg / 760 mm Hg
= 0.263 = 26.3%

Q2002 In gas chromatography, the separation of sample components is determined primarily by what physical property of substances?

1. Number of carbon atoms
2. Boiling point
3. Heat of vaporization
4. Ability to absorb UV

Ans(2). The boiling point is the determining factor within a group of compounds. Physical properties will effect the separability of the compounds.

Q2003 Spectrophotometry uses absorption in the ultraviolet, infrared, and visible wavelengths.

1. True
2. False
3. Depends upon water content of substance
4. Depends upon hydrocarbon content of substance

Ans(1). Spectrophotometry is a group of analytical techniques which utilize electromagnetic radiation to conduct qualitative or quantitative analysis of various substances.

Q2004 Strong oxidizing agents:

1. React violently with bases
2. Can cause serious burns
3. Shall be stored in steel containers
4. React violently in air

Ans(2). Oxidizers do not always react violently with bases nor can they be stored safely in steel containers. They can however produce severe burns.

Q2005 The density of an unknown ideal gas at 85ºC and 720 mm Hg pressure is 2.75 g/l. What is the gram molecular weight of this gas?

1. 0.32 g/mol
2. 3.08 g/mol
3. 30.8 g/mol
4. 85.4 g/mol

Ans(4). Temperature is 358° K. Pressure is 720 mm Hg/760 mm Hg/atm = 0.947 atm
Calculate the number of moles in 1 liter.
PV = nRT
n/V = P/RT
= 0.947 atm / (0.082 l-atm/K-mol * 358° K)
= 0.0322 mol/l
Since 0.0322 mole weighs 2.75 g, 1 mole weighs:
Weight = (2.75 g/l) / (0.0322 mol/l)
= 85.4 g/mol

Q2006 The elements of silane include:

1. Si, H
2. Cl, H, C, O
3. As, H
4. S, H

Ans(1). The molecular formula for silane, or silicon tetrahydride, is SiH_4. Therefore silicon and hydrogen are its elemental constituents.

Q2007 The pressure in a closed gas cylinder is doubled as a result of heating. If the original ambient temperature was 25ºC, what is the new temperature?

1. 50ºC
2. 149ºC
3. 323ºC
4. 596ºC

Ans(3).25º C = 298° K
$(P_1 V_1) / T_1 = (P_2 V_2) / T_2$
$(P_1 V_1) / 298 = 2 (P_2 V_2) / T_2 P_2$
$= 2(P_1 V_2) = V_1$ (it is a closed container)
$T_2 = (2 * P_1)(V_1)(298) / P_1 V_1$
= 596° K = 323ºC
(Note that P_1 and V_1 are algebraically eliminated. Also, remember to convert Kelvin back into Celsius.)

Q2008 An atmosphere-supplying respirator must supply a quality of breathing air which contains the normal amount of oxygen, no more than 5 mg/m^3 of condensed hydrocarbons, no more than 10 ppm of carbon monoxide, and other requirements. What Grade of air does this describe?

1. Grade A
2. Grade B
3. Grade C
4. Grade D

Ans(4).Grade D breathing air is required by OSHA and its quality is specified by the ANSI/CGA Commodity Specification for Air and re-stated by OSHA in the Respiratory Protection Standard. In addition to those requirements listed above, the air must also have no pronounced odor and have no more than 1,000 ppm of carbon dioxide. The oxygen concentration must be between 19.5% and 23.5% by volume. ANSI/CGA G-7.1-1989 29 CFR 1910.134 (i)(1)(ii)

Q2009 An air-purifying, half-face piece respirator has a protection factor of:

1. 1
2. 5
3. 10
4. 50

Ans(3).The half-face piece air purifying respirator has a protection factor of 10.

Q2010 Pressure demand units are approved for use in IDLH atmospheres. The approvals are for units that last 15, 30, and 60 minutes. There are also approved units with times of 3, 5, 10 minutes. These latter units are approved for:

1. Escape only
2. Escape and entry
3. Entry only
4. Not for industrial applications

Ans(1). The short duration pressure demand units having use times of 3, 5, and 10 minutes have been approved for escape only. They are not approved for entry into non-respirable atmospheres.

Q2011 The following are true regarding the TLV's for simple asphyxiants, EXCEPT:

1. They are inert gases and vapors without significant physiological effects
2. Employees should not be exposed above 5,000 ppm
3. Most do not have TLV's
4. Exposure limit is based on % of available oxygen

Ans(2). Simple asphyxiants are inert gases and vapors that, when present in high concentrations in air, act primarily by displacing available oxygen. 5,000 ppm is a high concentration that could be dangerous for flammable asphyxiants such as methane, but is not high enough to reduce the oxygen level from 20.5% to the minimal oxygen level of 18%.

Q2012 A disposable dust mask has a protection factor of approximately:

1. 2
2. 5
3. 10
4. 50

Ans(2). Disposable respirators such as dust masks have protection factors of 5. The protection factor is the ratio of the concentration outside the mask compared to the concentration inside the mask. Two (2) is less than most respirators, 10 is the protection factor for half mask cartridge respirators. and 50 is the protection factor of demand supplied air respirators.

Q2013 Pressure demand SCBA's have a protection factor of:

1. 10
2. 100
3. 1,000
4. 10,000

Ans(4). A pressure demand self-contained breathing apparatus has a protection factor of 10,000. The protection factor is the ratio of the concentration outside the mask compared to the concentration inside the mask.

Q2014 The least important consideration in evaluating the length of time an air-purifying respirator can be used is the:

1. Concentration of the contaminant
2. Duration of exposure
3. Type of cartridge used
4. Toxicity of the contaminant

Ans(4). Although the toxicity of the contaminant is an important consideration when selecting a respirator, the length of time the respirator can be used is determined by the concentration, duration, and type of cartridge.

Q2015 How many minutes of air do you have in SCBA emergency conditions?

1. 60
2. 30
3. Between 15 and 45
4. Significantly less than the rated number of minutes for the cylinder

Ans(4). Due to physical exertion, air is used very quickly in an emergency, significantly reducing the number of minutes of air delivered from an SCBA.

Q2016 A typical hydrocarbon emission rate for a stationary source coal combustion unit is 0.4 pounds hydrocarbon per ton of coal. What is the annual hydrocarbon emission rate from a power plant burning 300 tons of coal per day?

1. 15.7 tons
2. 17.9 tons
3. 21.9 tons
4. 34.3 tons

Ans(3).0.4 lbs/ton HC * 300 ton/day = 120 lbs/day
120 lbs/day * 365 days/yr = 43,800 lbs/yr
HC 43,800 lbs/yr / 2000 lbs/ton = 21.9 tons/yr HC

Q2017 A point source produces 1 pound of pollutant every hour. How many tons will be produced in one year assuming continuous 24-hour operation?

1. 2.2 tons
2. 2.4 tons
3. 3.6 tons
4. 4.4 tons

Ans(4). Total = 1 lb/hr * 24 hr/day * 7 days/wk * 52 wk/yr
= 8,736 lb/yr * 1 ton/2,000 lb
= 4.4 tons/yr

Q2018 Catalytic combustion of pollutants implies a _____ burn temperature.

1. Lower
2. Similar
3. Higher
4. Lower (if molecular weight is high)

Ans(1). Catalytic combustion technology was introduced to lower the temperature required to oxidize pollutants.

Q2019 What is the most important potential consequence of mixing different waste materials together?

1. Unexpected/unwanted chemical reactions
2. Increased cost and difficulty of waste characterization and treatment/disposal arrangements
3. Decreased feasibility of reuse or recycling
4. Injuries

Ans(4). The consequences of mixing incompatible chemicals together is severe from a health and safety perspective. From a disposal perspective, mixing of wastes can have a severe cost impact resulting from the Land Disposal Restrictions. Mixed wastes can require sequential treatment using different technologies at different facilities.

Q2020 The indicator organism used to measure water quality is:

1. Coliform bacteria
2. Hepatitis B
3. Polio virus
4. Streptococcus

Ans(1). The EPA Drinking Water Standard uses coliform bacteria as an indicator of microbial contamination.

Q2021 Equipment that monitors leachate is used with:

1. Air scrubbers
2. Hazardous waste disposal
3. Incinerators
4. Gas fired boilers

Ans(2). Leachate is a combination of groundwater or surface water that has infiltrated the hazardous solid waste and suspended solid matter, microbial waste products, heavy metals, and other chemical substances in a hazardous waste landfill. Leachate comes to the surface or percolates down to the groundwater. Analysis equipment is used to determine the types of heavy metals, chemicals, or biological agents found in the leachate.

Q2022 In waste parlance, the term "rubbish" and "garbage" have different meanings. Rubbish is that portion of waste not containing:

1. Cloth
2. Paper
3. Wood
4. Food

Ans(4). Garbage" refers to food waste while rubbish refers to non-hazardous non-food waste.

Q2023 Critical orifice flow is dependent upon upstream pressure:

1. Directly
2. Indirectly
3. As the square
4. Square root

Ans(1). The mass flow rate in a critical orifice is directly proportional to the upstream pressure.

Q2024 Critical pressure is defined as the:

1. Point where temperature inversion occurs
2. Absolute pressure at 473º K
3. Point where no liquid-vapor phase transition occurs
4. Pressure of freezing water

Ans(3).The critical pressure of a substance is that pressure at which no liquid-vapor phase transition occurs for any temperature.

Q2025 For the following decomposition reaction, how many liters of hydrogen would be released if 17 g of ammonia were decomposed? Assume ideal gas behavior at STP. 2 (NH_3) → N_2 + 3 (H_2)

1. 3.4 liters
2. 16.3 liters
3. 33.6 liters
4. 36.7 liters

Ans(3). One mole of a gas at STP occupies 22.4 liters. For every mole of ammonia ,3/2 or 1.5 moles, of hydrogen is released. 17 g of ammonia (NH_3) equals one mole, therefore, hydrogen = 22.4 l * 1.5 = 33.6 liters

Q2026 Given that the volume of a gas at 45ºC is 79.5 ml., what would the volume be at 0ºC? Use Charles' Law to determine the new volume.

1. 6.83 ml
2. 68.3 ml
3. 78.3 ml
4. 88.5 ml

Ans(2). Charles' law states that volume is proportional to absolute temperature (i.e., $V_2 = V_1(T_2 / T_1)$. In this way if the temperature rises volume increases. If temperature decreases the volume will decrease. Remember to always use absolute temperatures.
$V_2 = V_1 (T_2 / T_1)$
= 79.5 ml (273° K / 318° K)
= 68.3 ml

Q2027 The atomic weight of carbon is 12, oxygen is 16, and chlorine is 35. What is the atomic weight of a molecule of phosgene ($COCl_2$)?

1. 98
2. 63
3. 105
4. 89

Ans(1). 12 + 16 + 35 + 35 = 98

Q2028 Given the following meter readings in gallons per minute, what is the total number of gallons discharged in 6 hours?
0 - 3 hrs 12 gpm
3 - 6 hrs 15 gpm

1. 3,000 gal
2. 3,460 gal
3. 4,550 gal
4. 4,860 gal

Ans(4). You must convert the gpm reading to gallons per hour.
(12 gal/min * 60 min/1 hr) = 720 gal/hr
12 gpm = 720 gal/hr
15 gpm = 900 gal/hr
Total = 3 hrs at 900 gal/hr + 3 hrs at 720 gal/hr
= (3 hr * 900 gal/hr) + (3 hr * 720 gal/hr)
= 2,700 gal + 2,160 gal
= 4,860 gal

Q2029 How many carbon atoms does a benzene ring have?

1. 4
2. 5
3. 6
4. 8

Ans(3).A benzene ring has 6 carbon atoms.

Q2030 How many moles of sulfur are there in 10 g of the element? The atomic weight of sulfur is 32.

1. 0.3 mol
2. 3.0 mol
3. 30.0 mol
4. 32.0 mol

Ans(1). Knowing the atomic weight of sulfur you know that one mole weighs 32 grams (atomic weights have units of grams/mole).
Moles = 10 g / (32 g/mol) = 0.3 mol

Q2031 If a material is contaminated with impurities, the melting point will be _____ than that of the pure material. This property is often used to assess purity in the laboratory.

1. Higher
2. Lower
3. Same
4. Lower and broader

Ans(4). Impurities will lower AND broaden the melting point of a material. Melting points are often used in the laboratory as an easy method to roughly assess purity.

Q2032 Which of these conversions is NOT true?

1. 1 liter H_2O = 1,000 g H_2O at STP
2. 1 m^3 = 10,000 liters
3. 14.7 psi = 760 mm Hg = 760 torr
4. 1 liter= 0.001 m^3

Ans(2).To succeed on the examination, one must be very comfortable with unit conversions. Frequently, problems on the examination will be in various non- related units. 1 m^3 = 1,000 liters

Q2033 Why does water pressure increase with depth?

1. Water becomes more dense with depth
2. Water exerts more force per unit area with depth
3. The force of gravity increases with depth
4. Unit mass of water changes with depth

Ans(2). Pressure is defined as a force per unit area. As the weight of water increases with increasing depth, the pressure or force per unit area increases.

Q2034 Why would a lead oxide particle settle faster than a silica particle?

1. Shape
2. Size
3. Density
4. Diffusion

Ans(3). Settling velocity is directly proportional to particle density; hence, the lead particle will settle faster because it has a higher density.

Q2035 What is atomic mass?

1. Proton number
2. Neutron number
3. Total number of neutrons plus protons
4. Average mass of naturally occurring isotopes

Ans(3). The atomic mass is the sum of the number of protons and the number of neutrons in a particular isotope of an element to yield its atomic mass. Atomic weight, which is a non-integer number (usually), is obtained by averaging the masses of the naturally occurring isotopes of the element.

Q2036 Which of the following compounds is not a member of the chemical class of alkylbenzenes?

1. Toluene
2. Xylene
3. Benzo(a) pyrene
4. Cumene

Ans(3). Benzo(a) pyrene is polyaromatic hydrocarbon (PAH) consisting of several fused benzene rings. The acute inhalation toxicity of alkylbenzenes is sometimes manifested as depression. Unlike benzene, there is no substantiated data indicating that these compounds cause hematologic effects such as anemias or leukemias.

Q2037 A chemical has a vapor pressure of 0.50 mm Hg at 25ºC. What is the chemical's concentration (ppm) in air saturated with its vapor at 25ºC?

1. 10 ppm
2. 300 ppm
3. 660 ppm
4. 1,520 ppm

Ans(3). PPM = (0.50 mm Hg/760 mm Hg) x 10^6
ppm = 658 ppm
The concentration of a vapor is simply its partial pressure multiplied by a conversion of 10^6 ppm.

Q2038 How much benzene would have to be evaporated in a room 10 m * 10 m * 8 m to give a final concentration of 10 ppm at STP? (The specific gravity of benzene is 0.879. Its molecular weight is 78.)

1. 15.5 ml
2. 25.5 ml
3. 29.0 ml
4. 31.9 ml

Ans(3). Determine total volume of room:
V = 10 m * 10 m * 8 m = 800 m^3
Determine the mg/m^3 equivalent to 10 ppm:
mg/mp3 = (ppm * MW) / 24.45
= (10 * 78) / 24.45 = 31.9 mg/m^3
Determine the mass of benzene in the room:
Mass = 31.9 mg/m^3 * 800 m^3 = 25,520 mg = 25.52 g
Determine the volume associated with a mass of 25.52 g:
Volume = 25.52 g / 0.879 g/ml = 29.03 ml

Q2039 Which of the following equations is CORRECT?

1. $V_1T_1 = V_2T_2$
2. $V_1/P_1 = V_2/P_2$
3. PT = nRV
4. ppm = (mg/m^3 * 24.45) / MW (at 25ºC and 1 atm)

Ans(4). Only ppm = (mg/m^3 * 24.45) / MW (at 25ºC and 1 atm) is correct. This is the formula for converting mg/m^3 to ppm.

Q2040 Convert (100 oz/1,000 ft^3) to mg/l:

1. 0.1 mg/l
2. 10 mg/l
3. 100 mg/l
4. 1,000 mg/l

Ans(3). 100 oz / 1,000 ft^3 = 0.1 oz/ft^3
Perform a dimensional conversion:
Conv = 0.1 oz/ft^3 * 1 lb/16 oz * 454 g/lb * 1,000 mg/g * 0.03531 ft^3/l = 100 mg/l

Q2041 Given one mole of an ideal gas at a temperature of 0ºC, and at a pressure of 1 atm, calculate the molar volume that this gas will occupy:

1. 22.4 l/mol
2. 24.2 l/mol
3. 24.4 l/mol
4. 28.8 l/mol

Ans(1). PV = nRT
V = nRT / P
= (1 mol * 0.08205 l-atm/mol-K * 273° K) / 1 atm
= 22.4 l/mol

Q2042 Given one mole of an ideal gas at a temperature of 25ºC, and a pressure of 1 atm, calculate the volume occupied by the gas:

1. 22.4 l/mol
2. 24.5 l/mol
3. 25.4 l/mol
4. 24.5 ml/mol

Ans(2).PV = nRT
V = nRT / P
= (1 mol * 0.08205 l-atm/mol-K * 298° K) / 1 atm
= 24.46 l/mol

Q2043 What is the total pressure of a gaseous mixture with partial pressures of 75 mm Hg of SF_6, 45 mm Hg CH_4, 50 mm Hg CO, and 120 mm Hg of N_2?

1. 200 mm Hg
2. 400 mm Hg
3. 300 atm
4. 290 torr

Ans(4). The total pressure is a sum of all of the partial pressures.
P = 75 mm Hg + 120 mm Hg + 45 mm Hg + 50 mm Hg
= 290 mm Hg = 290 torr
Remember that 1 torr is equivalent to 1 mm Hg.

Q2044 What is the specific gravity of mercury, which is a liquid at 25ºC? (The density of mercury is 13.534 g/ml, and the density of water is 0.997 g/ml.)

1. 13.57 kg/m^3
2. 13.57
3. 13.57 mg/m^3
4. 13.57 g/ml

Ans(2). Specific gravity is the ratio of the density of a substance to that of a reference material. In the case of a liquid, the reference material is water.
SG (Hg) = ⟩ Hg / ⟩ H_20
= (13.534 g/ml) / (0.997 g/ml)
= 13.57
Note that there are no units to specific gravity.

Q2045 Ethyl alcohol has a molecular weight of 46.07 g/mol and a density of 0.785 g/ml at 25ºC. What is the volume occupied by 1 mole of ethanol at NTP?

1. 58.5 liters
2. 58.5 ml
3. 50 liters
4. 50 ml

Ans(2). ml/mol = 46.07 g/mol * 1 ml/0.785 g
= 58.5 ml/mol ml
= ml/mol * number of moles
= 58.5 ml/mol * 1 mol
= 58.5 ml

Q2046 What is the volume of a 0.76 meter high column of mercury if the cross-sectional area of the column is 1 square meter?

1. 0.076 m^3
2. 0.76 m^3
3. 1.14 m^3
4. 1.32 m^3

Ans(2).V = hA
h is the height and A is the cross-sectional area
V = 0.76 m(1 m^2)
= 0.76 m^3

Q2047 A Btu is defined as:

1. The amount of energy required to raise 1 kg of water 1ºF
2. 1/180 of the amount of energy required to raise the temperature of one pound of water from 32ºF to 212ºF
3. The amount of energy needed to metabolize 1 lb of fat
4. The amount of heat needed to raise the temperature of 1 lb of water from 32ºF to 212ºF

Ans(2). The amount of heat required to raise water is a specific amount and varies with temperature. 1/180 of the amount of energy required to raise the temperature of one pound of water from 32ºF to 212ºF accounts for the change of the heat capacity of water as its temperature changes, giving the average over the whole range of temperatures at which water is a liquid at 1 atm. The more common definition of a Btu is defined as the heat necessary to raise 1 lb of water 1ºF.

Q2048 Physical adsorption is characterized by:

1. Electromagnetic forces on media surface
2. Reaction with a media to form new compounds
3. Chemical reaction on the surface of a media
4. Penetration of a media by a chemical

Ans(1). Physical adsorption is characterized by electromagnetic forces attracting a chemical to a media. Chemical reaction on the surface of a media is the mechanism for chemical adsorption. Penetration of a media by a chemical is the mechanism for physical absorption.

Q2049 The volume of a gas at 45ºC is 79.5 ml. What is the volume at 0ºC? Use Charles' Law to determine the new volume.

1. 6.83 ml
2. 68.3 ml
3. 78.3 ml
4. 88.5 ml

Ans(2).Charles' Law states that volume is proportional to absolute temperature, i.e., $V_2 = V_1 (T_2/T_1)$. In this way, if the temperature rises, volume increases. If temperature decreases the volume will decrease. Remember to use absolute temperatures. $V_2 = 79.5$ ml(273° K/318° K) = 68.3 ml

Q2050 When the temperature is raised, the rate of reaction increases. What happens to endothermic reactions when the temperature is raised?

1. Increases
2. Decreases
3. No effect
4. Minor decrease as long as pressure is above atmospheric pressure

Ans(1).Increasing the temperature will increase the rate of any reaction, whether it is exothermic or endothermic because increasing temperature increases the movement of atoms/molecules. Increased movement means increased collisions, increased collisions means increased reactions.

Q2051 The major law of chemistry upon which visible and ultraviolet spectrophotometry is based is:

1. Fick's Law
2. Beer's Law
3. PV = nRT
4. $P_1 V_1 = P_2 V_2$

Ans(2). Spectrophotometry is based upon Beer's Law. Beer's Law states that A = a * b * c, where A is absorption, a is absorptivity, b is cell thickness, and c is concentration.

Q2052 50 g of chlorine (Cl_2) contains how many chlorine molecules? (The atomic weight of chlorine is 35.5 g/mol.)

1. 4.24×10^{23} molecules
2. 42.4×10^{23} molecules
3. 8.48×10^{23} molecules
4. 84.8×10^{23} molecules

Ans(1). Molecular weight of Cl_2 = 35.5 g/mol * 2 = 71 g/mol. This indicates that each mole of chlorine has a gram-molecular weight of 71 g/mol. The number of molecules in one mole of ANYTHING is 6.02×10^{23}. Therefore, Molecules = 50 g / (71 g/mol) * 6.02×10^{23} molecules/mol = 4.24×10^{23} molecules

Q2053 Molality refers to:

1. Number of moles of solute per liter of solution
2. Number of moles solute per 1,000 grams of solvent
3. Numbers of moles per gram of molecular weight
4. Number of grams of solute per mole of solution

Ans(2). Molality refers to the number of moles of a solute dissolved per 1,000 grams of a solvent. Number of moles of solute per liter of solution is molarity.

Q2054 What is the correct way to convert mg/m^3 to g/l?

1. Multiply mg/m^3 by 1,000 g/l per mg/m^3
2. Multiply mg/m^3 by 1 g/l per mg/m^3
3. Multiply mg/m^3 by 10 g/l per mg/m^3
4. Multiply mg/m^3 by 10^{-6} g/l per mg/m^3

Ans(4). Since there are 1,000 milligrams in a gram and 1,000 liters in a cubic meter, the conversion is:
$mg/m^3 = mg/m^3$(1 g/ 1,000 mg)(1 m 3/1,000 liters)
$= 10^{-6}$ g/l

Q2055 In a titration, 30 ml of a base was used to neutralize 25 ml of a 2.0 N acid. What is the normality of the base?

1. 1.67
2. 2.00
3. 2.33
4. 3.00

Ans(1). Volume acid * Normality acid = Volume base * Normality base
25 ml * 2.0 = 30 ml * Normality base

Q2056 The volatility of a chemical refers to the:

1. Capacity of a liquid to form vapors
2. Degree of flammability of a chemical
3. Exothermic reaction of a liquid
4. Endothermic reaction of a liquid

Ans(1). Volatility refers to the capacity of a liquid to form vapors, which can be measured through its vapor pressure. Flammability of a liquid is increased if it is more volatile, but non-flammable liquids can be volatile also. Exothermic and endothermic reactions are not related to volatility.

Q2057 The equilibrium between molecules of gas in the atmosphere and molecules of the same gas in solution is defined by:

1. Faraday's Law
2. Henry's Law
3. Friedel-Crafts reaction
4. Plank's Law

Ans(2). Henry's Law states that the solubility of a gas in a liquid is proportional to the partial pressure of that gas-liquid interface.

Q2058 In the Ideal Gas Law, PV = nRT, T is expressed in:

1. Fahrenheit
2. Celsius
3. Centigrade
4. Kelvin or Rankine

Ans(4). In the Ideal Gas Law, the temperature is expressed in absolute temperature. The Kelvin scale and Rankine scale are absolute temperature scales.

Q2059 A container holds 50 ml of toluene with a specific gravity is 0.87 g/ml. What is the mass of toluene?

1. 13.2 g
2. 23.5 g
3. 43.5 g
4. 66.6 g

Ans(3). Specific gravity denotes the ratio of the mass of a substance to that of an equal volume of a reference substance. Specific gravity calculations are used to determine the mass of a volume of liquid.
50 ml of toluene * 0.87 g/ml = 43.5 g toluene

Q2060 How many grams of acid are in 500 ml of a 3 molar sulfuric acid solution?

1. 47 g
2. 98 g
3. 147 g
4. 196 g

Ans(3).Molar solution means mol/l. A 3 molar solution of sulfuric acid therefore contains 3 moles of sulfuric acid per liter.
Molecular weight of H_2SO_4 = 98 g/mol Grams
= 98 g/mol * 3 mol/l x 0.5 l
= 147 g

Q2061 How many moles of acid are in 500 ml of a 3 molar sulfuric acid solution?

1. 1.5 mol
2. 2.0 mol
3. 2.5 mol
4. 3.0 mol

Ans(1).Molar solution means mol/l. A 3 molar solution of sulfuric acid therefore contains 3 moles of sulfuric acid per liter.
500 ml = 0.5 liters
Moles = 3 mol/l * 0.5 liters
= 1.5 mol

Q2062 An oxygen tank is filled with oxygen at an absolute pressure of 1,000 pounds per square inch at 17ºC. The internal volume of the tank is 1 cubic foot. The oxygen is used in high-altitude safety simulations at an absolute pressure of 10 pounds per square inch at -27ºC. How many cubic feet of oxygen can be supplied by the tank under these conditions?

1. 70 ft^3
2. 75 ft^3
3. 80 ft^3
4. 85 ft^3

Ans(4).Use the Ideal Gas Law: $(P_1V_1) / T_1 = (P_2 V_2) / T_2$
Remember to use absolute temperatures.
(1,000 psi * 1 ft^3) / 290° K = (10 psi * V_2) / 246° K V_2 = 84.8 ft^3

Q2063 What is the pH of a solution with a concentration of 10^{-4} mol/l of hydrogen ions?

1. 3
2. 4
3. 9
4. 10

Ans(2). Use the formula: pH = -log [hydrogen ion concentration]
pH = -log [1 x 10^{-4}] pH
= -(-4) log 1 pH = 4

Q2064 A liquid with a pH of 7 is:

1. Basic
2. Acidic
3. Neutral
4. Alkaline

Ans(3). A liquid with a pH of 7 is neutral.

Q2065 The explosion of an atomic bomb exemplifies the conservation of energy principle in which:

1. Energy is converted into matter
2. Energy is converted into an explosion
3. Energy begets more energy by means of a nuclear explosion
4. Matter is converted into energy

Ans(4).The explosion of an atomic bomb involves the conversion of uranium atoms to other atoms with the evolution of energy in the process. The energy is generated from the conversion of a given mass of the starting atoms into energy.

Q2066 Due to environmental concerns, what major type of fire suppression agent has been removed from production and replaced with halocarbon and inert gas agents?

1. Halon
2. Carbon dioxide
3. Dry chemical
4. Nitrogen

Ans(1). Halon is an ozone depleting agent that was banned by the Montreal Protocol. Carbon dioxide, dry chemicals, and nitrogen are currently used for fire suppression. Nitrogen is used as an inerting or suppression agent over oil baths.

Q2067 One potential by-product of water chlorination that is NOT beneficial is:

1. Destruction of E. coli
2. Lowered turbidity
3. Presence of trihalomethanes
4. Destruction of pathogens

Ans(3). The formation of trihalomethanes by water chlorination is an adverse byproduct. Chloroform, one of the trihalomethanes, is a suspected human carcinogen.

Q2068 Three layers of the skin include:

1. Hair follicles, capillaries, muscles
2. Epidermis, dermis, subcutaneous tissue
3. Id, ego, superego
4. Cuticles, capillaries, muscles

Ans(2). The 3 layers of the skin are the epidermis (outer layer), the dermis (true skin), and the subcutaneous tissue.

Q2069 The most important function of these micro-organisms in the environment is to break down the cellulose in wood and other plant materials.

1. Fungi
2. Bacteria
3. Algae
4. Amoeba

Ans(1). Fungi generate an enzyme called cellulase which breaks down insoluble cellulose to soluble carbohydrates.

Q2070 In heat stress science, metabolism is assessed as:

1. Negative
2. Positive
3. Can be both positive and negative
4. Negligible

Ans(2).In heat stress science, the heat gained by the body through metabolism is positive.

Q2071 In which part of the body is the main thermoregulation activity handled?

1. Lungs
2. Heart
3. Skin
4. Hair

Ans(3). Skin is the site of most thermoregulation activities including sweating, and conductive and radiant heat transfer.

Q2072 The primary constituent of Hopcalite is?

1. Lime
2. Manganese dioxide
3. Zinc
4. Borate

Ans(2). Hopcalite is a specially prepared mixture of manganese dioxide with other oxides used for oxidizing CO.

Q2073 The amount of water a public water system must provide includes all the following EXCEPT:

1. Domestic consumption
2. Leakage in the system
3. Industrial consumption
4. Fire fighting service

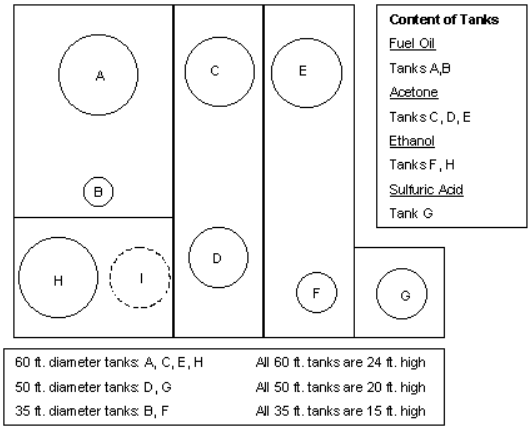
Ans(2). The amount includes all items except the system leakage which would be included in the domestic and industrial figures.

Q2074 An example of a mechanical collector for pollutants is:

1. A cyclone
2. An ESP
3. An afterburner
4. An incinerator

Ans(1). A cyclone collects pollutants by placing the incoming pollutant laden air into a vortex where the particulate matter drops out.

Q2075



Materials	Separation Distance
Acetone and . . .	
Acetone	50 ft.
Ethanol	50 ft.
Fuel Oil	40 ft.
Sulfuric Acid	80 ft.
Ethanol and . . .	
Acetone	50 ft.
Ethanol	50 ft.
Fuel Oil	40 ft.
Sulfuric Acid	80 ft.
Fuel Oil and . . .	
Acetone	40 ft.
Ethanol	40 ft.
Fuel Oil	30 ft.
Sulfuric Acid	80 ft.
Sulfuric Acid and . . .	
Acetone	80 ft.
Ethanol	80 ft.
Fuel Oil	80 ft.
Sulfuric Acid	55 ft.

A tank farm is being designed and will be located in a geographical location that receives extensive rainfall. The tank farm will house four hazardous materials: fuel oil, ethanol, acetone, and sulfuric acid. To ensure that corporate safety and environmental protection policies are followed, you have been asked to provide input on the layout of the containment dikes.
Because of the soil, no dike can be higher than 4.5 ft. Due to the potential for heavy rains, company policy requires a 25% overage be designed into the capacity of any given diked volume.
A new tank (Tank I) is scheduled to be constructed after this tank farm is built and will be placed adjacent to Tank H. The piping drawings indicate that the horizontal centerlines of Tank H and Tank I line up. Tank I will be a duplicate of Tank H and will contain the same material.
With respect to the graphic, the dimensions of the diked area containing Tank H are 190 ft horizontally x 140 ft vertically. If the design requires considering the anticipated addition of Tank I adjacent to Tank H, what is the minimum dike height surrounding Tank H?

1. 3.6 ft
2. 3.8 ft
3. 4.1 ft
4. 4.5 ft

Ans(1). Answer this question one step at a time.

Step 1: Find the capacity of either Tank H or Tank I (they are identical). Dikes must be able to contain a release of the largest tank in the diked area. Both Tank H and Tank I are 60 ft diameter tanks that are 24 ft high. Assume that the capacity of the tank is the same as the volume of a cylinder. Its volume is therefore:
$V = \pi[(60\ ft)^2 / 4]\ 24\ ft$
$= 67{,}858\ ft^3$

Step 2: The next step is to determine the required capacity of the diked area. The dike must contain a complete loss of either tank plus a rainfall overage of 25%. Therefore, the actual capacity of the dike is 25% higher than the volume of either tank.
Total Volume = $1.25 * 67{,}858\ ft^3$

= 84,823 ft^3

Step 3: Unfortunately, we cannot use the volume occupied by one of the tanks in the diked area. We DO assume that the volume occupied by the ruptured/leaking tank is available, though. We do not know the height of the dike, but we DO know the surface area of either tank. We will use this number shortly. Calculate the unavailable surface area of either tank.
Tank Surface Area = □ [(60 ft)^2 / 4]
= 2,827 ft^2

Step 4: Based on the dimensions given for the dike surrounding Tank H, we can determine the existing surface area of the dike and subtract the additional surface area made unavailable by the other tank.
Existing Surface Area = 190 ft * 140 ft
= 26,600 ft^2
Net Surface Area = Existing Surface Area - Tank Surface Area
= 26,600 ft^2 - 2,827 ft^2
= 23,773 ft^2

Step 5: Calculate the height of the dike by dividing the Total Volume by the Net Surface Area.
Dike Height = 84,823 ft^3 / 23,773 ft^2
= 3.57 ft

Q2076 The typical "scrubber" used in a factory for acid vapor emissions control is:

1. Filter
2. Electrostatic precipitator
3. Cyclone
4. Wet collector

Ans(4). A wet collector controls emissions through absorption of the air stream with a caustic liquid.

Q2077 Secondary standards as regards to drinking water are related to:

1. Health data
2. Aesthetics
3. Cancer risk
4. Reproductive hazard risk

Ans(2). Secondary standards for drinking water are related to aesthetic values that are more socially related.

Q2078 The removal of insoluble matter from water is referred to as:

1. Primary waste treatment
2. Secondary waste treatment
3. Tertiary waste treatment
4. Activated sludge process

Ans(1).Primary waste treatment removes insoluble matter through screening, removal of grit, and primary sedimentation

Q2079 Acronyms are used extensively in the various regulatory programs as a form of shorthand. Which of the following acronyms is not defined by the RCRA program?

1. TCLP
2. RQ
3. SQG
4. LQG

Ans(2). RQ, standing for Reportable Quantity, is defined under the CERCLA program. Please note, however, that hazardous wastes are by definition a subset of hazardous substances, and RQs have been assigned to each listed hazardous waste and each characteristic of hazardous waste. TCLP, SQG, and LQG are acronyms for RCRA defined terms of toxicity characteristic leaching procedure, small quantity generator, and large quantity generator, respectively.

Q2080 The toxic chemical which caused the Bhopal disaster had the acronym MIC. This stands for:

1. Methyl iso-cyanate
2. Methyl iso-carbamate
3. Methyl iso-chromate
4. Methyl iso-oxylate

Ans(1). Methyl iso-cyanate was the toxic chemical released during the Bhopal disaster.

Q2081 In situations involving the transfer of industrial property, the buyer should:

1. Determine whether there are any state laws regarding the environmental aspects of real property transfers
2. Perform business and environmental "due diligence" audits
3. Determine whether the necessary environmental permits have been obtained, and whether the facility is in compliance with permit terms and conditions. Identify any limitations on permit transfers and the appropriate procedures
4. All of the above

Ans(4). All of the above activities are prudent in situations involving the transfer of industrial properties due to the risks, costs, and liabilities under common law, state environmental statutes, and federal environmental statutes, especially CERCLA (Comprehensive Environmental Response, Compensation, and Liability Act) or "Superfund".

Q2082 Maximize the useful lifetime of process solutions

1. Maximize the useful lifetime of process solutions
2. Minimize inadvertent releases or wastage
3. Incorporate preventative maintenance and spill containment measures
4. All of the above

Ans(4). Good operating practices as waste minimization measures can be implemented to achieve any or all of the above listed objectives.

Q2083 One key factor in management success is Crisis Management. Which incident is an example of exemplary crisis management?

1. Union Carbide and Bhopal
2. Union Carbide and Institute, West Virginia
3. Johnson & Johnson and Tylenol
4. CH2M Hill and the Milwaukee tunnel collapse

Ans(3).Johnson & Johnson's handling of the Tylenol scare is considered an excellent example of effectively dealing with a major crisis.

Q2084 A flood destroys your company's operations facility. After the emergency management issues are addressed, your company implements several plans to:

1) recover critical files and information that had been stored offsite,
2) set up a temporary facility from which operations can be conducted, and
3) inform customers of the situation and how they will be served.

These plans are among examples of a comprehensive loss control activity called:

1. Emergency management/emergency response
2. Situational awareness
3. Business interruption recovery/business continuity
4. Inland marine preparedness

Ans(3).Business continuity planning is a key part of a loss control program. Such plans should include recovering corporate information, setting up operations, and financing temporary operations until a new facility can be commissioned. Depending upon the risk of a natural disaster, some companies purchase business interruption insurance to help finance operations.

Q2085 Waste minimization measures are driven by:

1. Statutory and regulatory requirements
2. Economic considerations
3. Legal liabilities
4. All of the above

Ans(4). All of the above listed considerations drive waste minimization/pollution prevention efforts. The Pollution Prevention Act of 1990, while not regulatory in nature, is evidence of Congressional interest in this area. The Hazardous and Solid Waste Amendments to RCRA resulted in waste minimization certifications being placed on manifests, and requirements for discussion of waste minimization efforts and achievements being added to biennial reports. Many states have their own waste minimization requirements.

Q2086 PRP is an acronym for:

1. Perpetually Responsible Party
2. Potentially Responsible Party
3. Principal's Responsibility Provision
4. Person Receiving Prison (Sentence)

Ans(2). PRP has become a common term in the discussion of responsibility for environmental contamination. In the United States, the EPA allows the use of the word "potentially" in front of the term "responsible party" in the CERCLA context.

Q2087 Joint and several liability defines a scope of liability in the United States. The practical meaning of joint and several liability is best described as:

1. All responsible parties must jointly contribute to clean-up costs, to the extent of their volumetric share of the wastes at the site
2. Each and every responsible party could be held individually liable for the entire cost of the clean-up
3. NPL sites where there are multiple responsible parties must act jointly, through generator committees, to severally apportion costs amongst themselves
4. EPA must allocate costs among responsible parties in a manner that reflects both the volume and toxicity of their wastes, as well as the difficulty of clean-up

Ans(2). The liability provisions under Section 107 of CERCLA are strict, joint and several, and retroactive in application.

Q2088 Strict liability can refer to the standard of liability in the United States. The practical implications of strict liability in cases of environmental contamination are best described as:

1. The liability applies regardless of fault, and the government does not have to prove intent or negligence on the part of the responsible party
2. Compliance with laws and regulations in effect at the time of disposal does not matter
3. The government does not need to prove that a responsible party's waste at the site actually was released or caused harm
4. All of the above

Ans(3). The liability provisions of CERCLA are found at Section 107, and are strict, joint and several, and retroactive.